DGCA CPL / ATPL STUDY NOTES • RADIO NAVIGATION • CH 10

✈ Chapter 10: MLS

Microwave Landing System

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1. Introduction & ILS Disadvantages

The Microwave Landing System (MLS) was designed to replace ILS as the ICAO standard precision approach system, overcoming ILS limitations while providing greater flexibility. However, few MLS installations exist and they co-exist with ILS.

⚠ Why ILS Needed Replacing — ILS Disadvantages

- Only **40 channels** available worldwide
- Fixed and narrow azimuth/glide slope beams → aircraft must be sequenced → landing delays
- No special procedures for helicopters, slower aircraft, STOL aircraft
- Cannot be sited in hilly terrain; requires large flat cleared land
- Vehicles, taxiing aircraft and buildings must be kept well clear to prevent beam bending

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2. The MLS System

- The MLS has a built-in DME.
- MLS is compatible with conventional localizer and glide path instruments, EFIS, auto-pilot systems and area navigation equipment.
- MLS gives positive automatic landing indications plus definite and continuous on/off flag indications for the localizer and glide slope needles.
- The identification prefix for the MLS is an 'M' followed by two letters.
- The aim is for all MLS equipped aircraft to operate to CAT III criteria. *Figures 10.1, 10.2 and 10.3* below show some of these features.

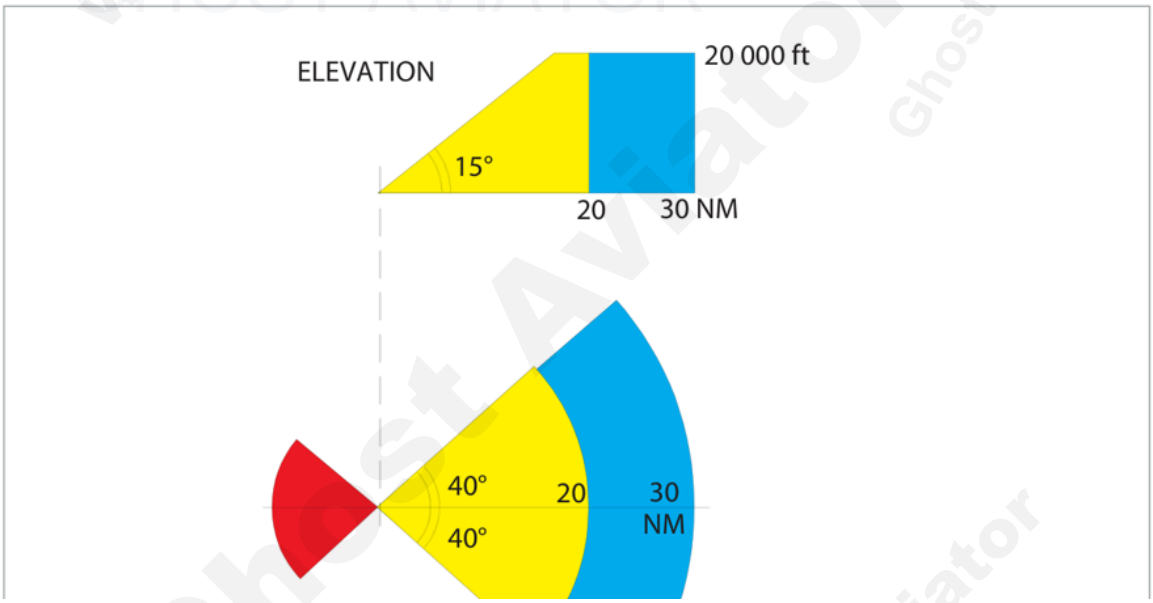


Fig 10.1: MLS coverage — ±40° azimuth, 20–30 NM range, up to 20,000 ft

MLS Key Features

Parameter	MLS Value	vs ILS
Channels	**200**	ILS: 40
Frequency band	**SHF: 5031–5090.7 MHz**	ILS: VHF/UHF
Azimuth coverage	**±40°**	ILS: ±35° (limited)
Glide slope range	**0.9° to 20°** selectable	ILS: fixed ~3°
Usable range (UK)	**20 NM** (up to 30 NM)	ILS: 25 NM/±10°

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Max coverage height	**20,000 ft**	ILS: variable
Back course	Secondary system ±20°, 15° elevation, 10 NM, 10,000 ft	ILS: problematic
DME	Built-in DME	ILS: optional
Ident prefix	**M** + two letters	ILS: I + letters
Aim	CAT III operations	ILS: CAT I/II/III

✓ MLS Advantages Over ILS

- SHF frequency: sitable in hilly terrain; scanning beam avoids reflections from vehicles/buildings
- 200 channels vs 40 for ILS → less frequency congestion
- Selectable approach paths (curved/segmented) → noise abatement procedures possible
- Pilots choose own approach path within coverage → better runway utilisation
- Compatible with EFIS, autopilot, RNAV and ILS-type displays
- Continuous on/off flag indications

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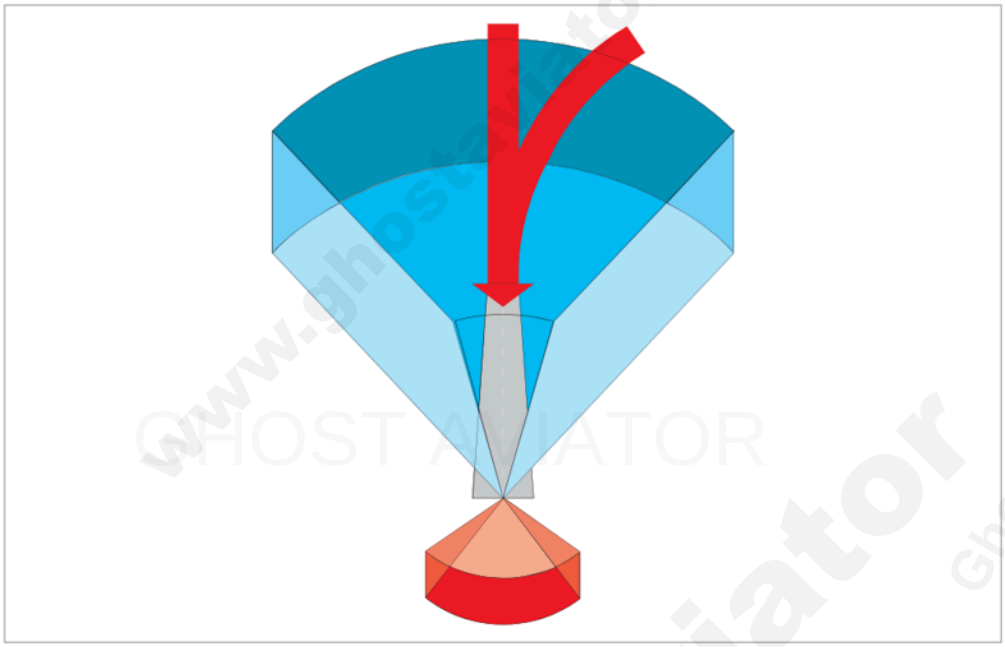


Figure 10.2 Approach Coverage Volume

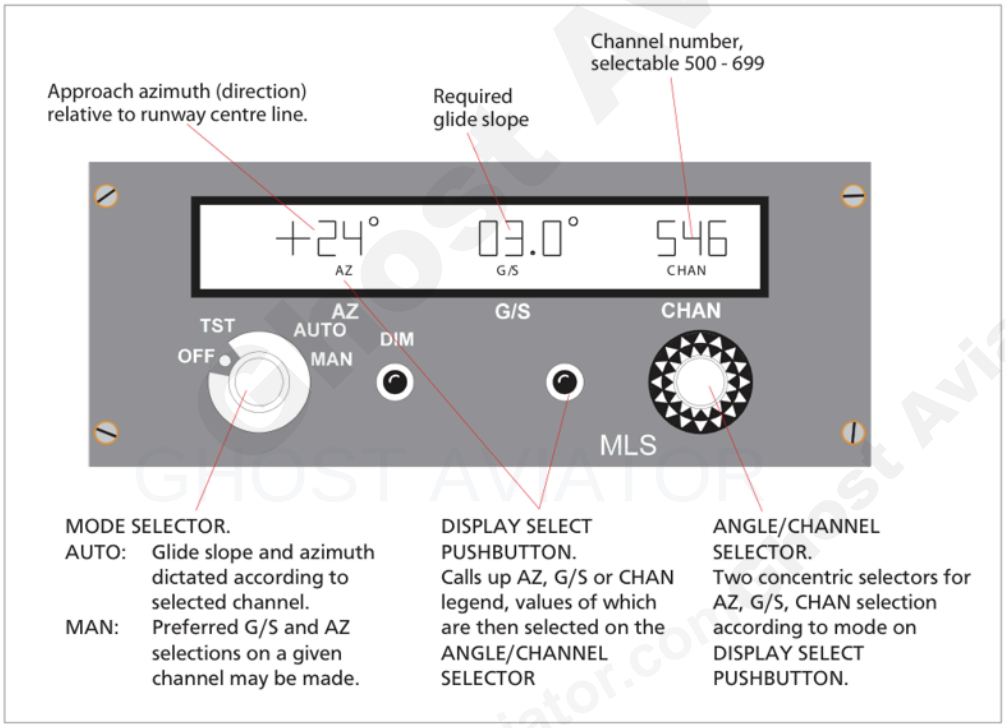


Figure 10.3 Typical MLS Flight Deck Control Panel

Fig 10.2: MLS approach coverage volume and Fig 10.3: Flight deck control panel

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3. Principle of Operation

MLS uses **Time Division Multiplexing (TDM)** — only one frequency per channel; all ground equipment transmissions are synchronized to avoid interference.

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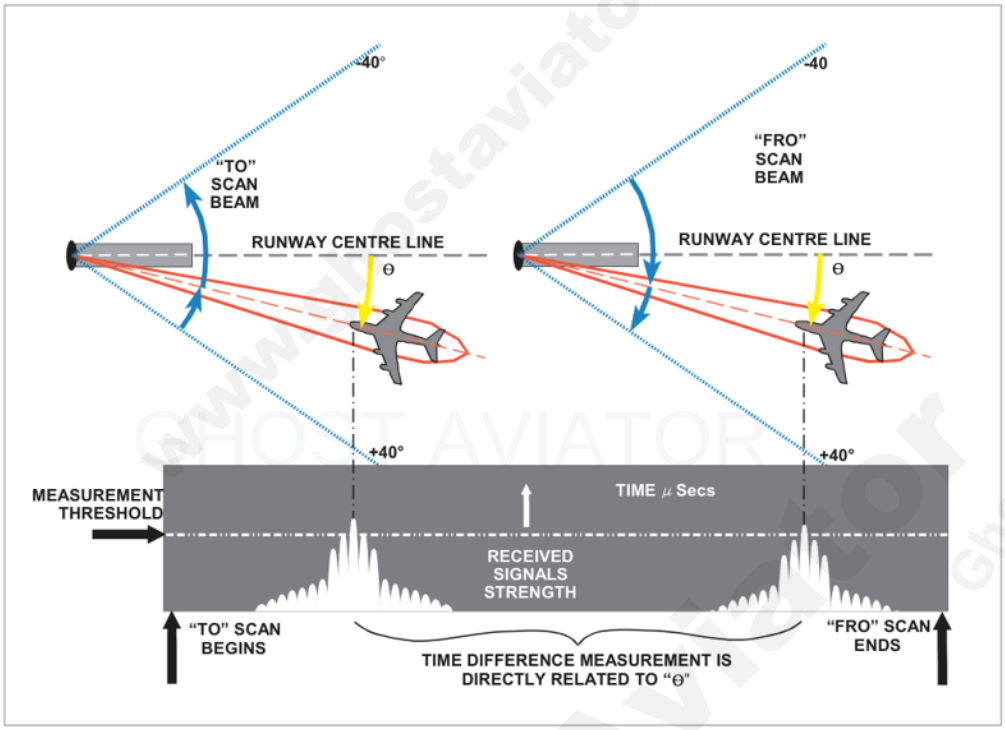


Figure 10.4

Fig 10.4: TO and FRO scanning beam — time difference proportional to aircraft angular position

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Time Referenced Scanning Beam (TRSB)

Measurement	How It Works
Azimuth	Beam sweeps “TO” then “FRO” across the scan sector. Aircraft measures the time interval between the two receptions. Time interval = angular position relative to runway centreline (QDM).
Elevation	Second beam sweeps up then down. Aircraft measures time difference between up/down pulses = angular position above horizontal = selected glide slope.
Range	Built-in DME (DME/P for Cat II/III — accurate to within **100 ft**).

Other MLS Components

- **Back Azimuth**

  : Go-around/departure guidance ±20° up to 15° elevation, 10 NM, 10,000 ft
- **Flare**

  : Standard provides for flare elevation (not intended for future implementation)
- **Auxiliary data**

  : Station ident, system condition, runway condition, weather information

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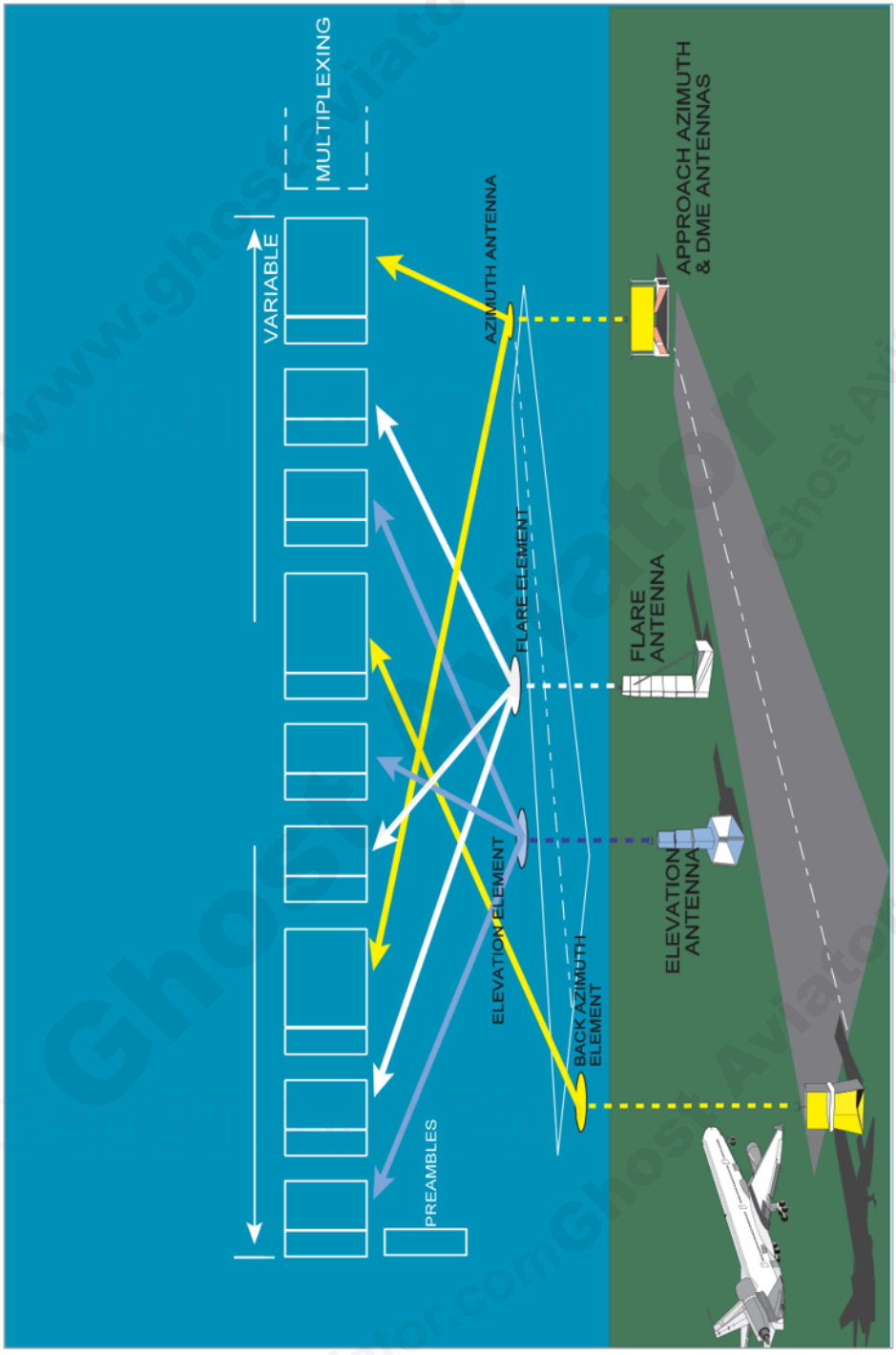


Figure 10.5 MLS component site

Fig 10.5: MLS component site layout

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4. Airborne Equipment

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Airborne Equipment

The airborne equipment is designed to continuously display the position of the aircraft in relation to the preselected course and glide path along with distance information during approach as well as during departure.

Display

The display consists of two cross bars similar to an ILS display except that the indications are given relative to the selective course. It is possible to program the computer to give segmented approaches and curved approaches for which a DME-P must be installed on the ground.

Control Unit

In order to receive ILS, MLS and GPS transmissions, aircraft are equipped with multi-mode receivers and a combined control unit for ease of use by the flight crew. An example of such a control unit is shown at *Figure 10.6*.



Figure 10.6 MLS control panel

Fig 10.6: MLS control panel — mode selector, display select, angle/channel selector

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MLS Airborne Equipment Features

- Continuously displays aircraft position relative to selected course and glide path + distance
- Two cross-bars similar to ILS CDI — indications relative to **selected** course (not fixed runway QDM)
- Can programme segmented and curved approaches (DME/P required on ground)
- **Multi-mode receivers**: handle ILS, MLS and GPS from single combined control unit
- Control unit allows selection of: AZ (azimuth angle), G/S (glide slope angle), CHAN (channel 500–699)
- Mode selector: AUTO (G/S and AZ auto per channel) or MAN (preferred G/S and AZ selection)

5. MLS Summary

Parameter	Details
Purpose	Precision approach system; designed to replace ILS
Frequency band	SHF: 5031–5090.7 MHz
Channels	200
Principle	TRSB (Time Referenced Scanning Beam) — time between TO/FRO sweeps
Azimuth coverage	±40° of runway QDM
Glide slope	0.9° to 20° selectable
Range (UK)	20 NM; up to 30 NM
Range measurement	Built-in DME (DME/P for Cat II/III: ±100 ft)
Ident	M + 2 letters
Display	Cross-bars (same as ILS CDI); relative to selected course
Target category	CAT III

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6. Practice Question

Q1. The coverage of the Microwave Landing System in the UK extends to ___ up to a height of ___ and ___ either side of the on-course line.

(a) 20 NM, 20,000 ft, 40 degrees

(b) 35 NM, 5,000 ft, 40 degrees

(c) 35 NM, 5,000 ft, 20 degrees

(d) 17 NM, 2,000 ft, 35 degrees

Answer: (a)

MLS in the UK: range **20 NM**, height **20,000 ft**, azimuth **±40°** of runway centreline.

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